Name_____ Date_____ Period_____

**LESSON 7-6 · HOMEWORK PACKET · SHEET 1 OF 2**

# Distance Between Points on a Grid

Maryland Standard 6.NS.C.8 · Learning target – I can find the distance between two points that share a coordinate.

**REMEMBER**

- **Same x-coordinate**: the points are on a vertical line. Use the y-coordinates.
- **Same y-coordinate**: the points are on a horizontal line. Use the x-coordinates.
- Opposite signs: **add** the absolute values. Same signs: **subtract** them.

**WORKED EXAMPLE**

Find the distance from (3, −4) to (3, 5).

**Step 1** Same x, so use the y-coordinates −4 and 5.

**Step 2** Opposite signs, so add: 4 + 5 = 9.

**Answer** **9 units**

## PRACTICE

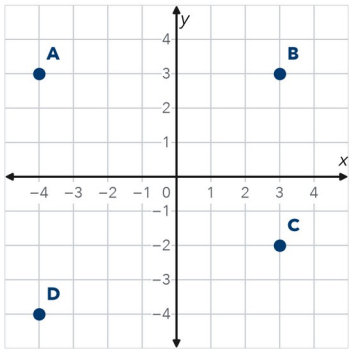
*show your work · label units*

**1.** Use the grid. Find each distance.

**a.** A to B _____ units

**b.** B to C _____ units

**c.** A to D _____ units




**2.** Find the distance between each pair of points.

**a.** (5, 6) and (5, −2) ____

**b.** (−7, 1) and (−2, 1) ____

**c.** (−3, −4) and (6, −4) ____

**d.** (0, −8) and (0, 3) ____

**e.** (−9, −2) and (−9, −7) ____

**f.** (2.5, 3) and (−4, 3) ____

**g.** (−1, −6) and (−1, 4) ____

**h.** (−8, 5) and (−3.5, 5) ____

**3.** **City Blocks.** Kim walks straight from (−6, −3) to (−6, 5). Each unit is 1 block. How far does she walk?

**Answer**_____ blocks

**4.** **Missing Point.** Point P is at (2, −3). Point Q is 6 units to the left of P. Name point Q.

**Answer**_____

**5.** **Find the Error.** Eli says the distance from (1, −6) to (1, −2) is 8 because 6 + 2 = 8. Explain his mistake and find the correct distance.

_____

_____

| **CHECK YOURSELF** | ☐ I can do this on my own | ☐ I need a little help | ☐ I have a question about #___ |
|---|---|---|---|

**AT HOME** With someone at home, count the blocks between two places on one street.

Name_____ Date_____ Period_____

**LESSON 7-6 · HOMEWORK PACKET · SHEET 2 OF 2**

# Comparing Distances on the Plane

Maryland Standard 6.NS.C.8 · Learning target – I can decide whether to add or subtract to find a distance, then compare two distances.

**REMEMBER**

- Find **each** distance first. Then compare.
- Different signs: **add**. Same sign: **subtract**.
- A distance is never negative.

**WORKED EXAMPLE**

Which is longer: (−2, 4) to (5, 4), or (1, −3) to (1, 5)?

**Step 1** First: same y, use x: 2 + 5 = 7 units.

**Step 2** Second: same x, use y: 3 + 5 = 8 units. 8 > 7.

**Answer** **the second one**

## PRACTICE

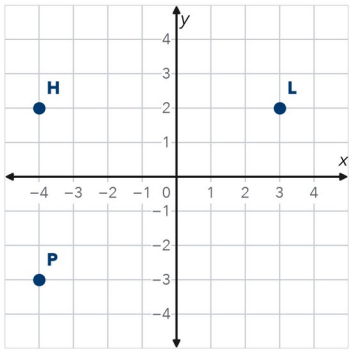
*show your work · label units*

**1.** **Neighborhood.** Each unit is 1 block. Which is closer to Home: the Library or the Park?

**a.** H to L _____ blocks

**b.** H to P _____ blocks

**c.** Which is closer? By how many blocks? _____




**2.** Find each distance.

**a.** (−8, 1) to (−3, 1) ____

**b.** (4, −6) to (4, 7) ____

**c.** (−6, −7) to (−6, −1) ____

**d.** (−2.5, 3) to (4.5, 3) ____

**3.** **Runners.** Ana runs from (−5, −1) to (6, −1). Jay runs from (2, 4) to (2, −6). Who runs farther, and by how many units?

**Answer**_____

**4.** **Reasoning.** For (−3, 7) to (−3, −2), do you add or subtract? Explain, then find the distance.

_____

_____

## TEST PRACTICE

*choose the best answer*

**5.** Segment JK runs from (−4, 2) to (6, 2). Segment MN runs from (1, −3) to (1, 5). Which is longer?

**A** JK, by 2 units **B** MN, by 2 units **C** neither **D** JK, by 8 units

**6.** Point A is at (−2, −5). Point B is at (−2, 3). What is the distance from A to B?

**A** 2 units **B** 3 units **C** 8 units **D** −8 units

**CHECK YOURSELF** ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #___

**AT HOME** With someone at home, pick two places on one street. Which is closer to your home? By how many blocks?